Option C – The Physics of Sports

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)	Any valid moment of inertia equation used i.e. $I = kmr^2$ (1) Valid conclusion based on value of moment of inertia of C with all values for A and B also considered (1)			2	2	1	
		(ii)	Angular momentum $L = I\omega$ used (1) Angular momentum = 22 [kg m^2 s^{-1}] (1)	1	1		2	1	
	(b)	(i)	Angular acceleration = $\frac{\text{change in angular velocity}}{\text{time used}}$ (1) Angular acceleration = 1 256 [rad s^{-2}] (1) Moment of inertia = 7.35×10^{-5} [kg m^2] (1) Torque = 0.099 [N m] (1)	1	1 1 1		4	3	
		(ii)	Rotational kinetic energy $= \frac{1}{2}I\omega^2$ used (1) Rotational KE = 2.5 [J] (**ecf** on I and angular velocity) (1)	1	1		2	2	
		(iii)	Ball has linear and rotational KE or recall $F = \frac{mv-mu}{t}$ (1) Energy has to be dissipated or the glove will <u>increase time</u> of contact when being caught (1) Reduce the effect of friction on the hand (1)	1	1 1		3		